Diet Optimization

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Introduction

- Unawareness about overall nutrient consumption
- Focus on balancing a person's daily nutritional intake by optimizing their fourth meal of the day
- Dataset:
  - Anuvaad Indian Nutrition Databank
  - Indian Council of Medical Research, MoFHW, Gov of India

Plan & Problem

- Followed PPDAC cycle
- Shannon entropy (motivation): This measures the nutritional diversity or "evenness" of a food item.

$$H = -\sum_{i=1}^{k} p_i log(p_i)$$

  It considers a single distribution.
- Kullback–Leibler Divergence: This measures how much a specific food's nutrient profile (p) diverges from a target profile (r= RDA - Consumed; in proportions)

$$D_{KL}(p \parallel r) = \sum_{i=1}^{k} p_i \log \left(\frac{p_i}{r_i} \right)$$

- p_i and r_i satisfy the three Kolgomorov's axioms of probability
- Returns the optimum nutrient proportions for the fourth meal given the consumed meals

Data

- Anuvaad Indian Nutrition Databank: Contains 1014 food items (mostly Indian, from from the US and the UK); 40 nutrients (per serving, per 100 g); food_name, primary_source
- Indian Council of Medical Research
  1. Common nutrients: magnesium, iron, zinc, B3 (niacin), B6, B9 (folate), vit C, vit A, vit D
  2. Body type (six out of twenty-three): age group, physical activity, gender
- Unavailability of data for other body types

Analysis

- We have used KL Divergence as our main tool for our Analysis.
- In the KL Divergence,
  - Target Distribution(R): $(r_1, r_2, ..., r_9)$, where $r_i = \frac{t_i}{\sum_9 t_j}$. Here, $t_i = |RDA_i - Consumed_i|$ for i^{th} nutrient.
  - True Distribution(X_j): $(x_{j1}, x_{j2}, ..., x_{j9})$, where $x_{ji} =$ proportion of i_{th} nutrient for j_{th} food item.
- M and N are the TUL and EAR proportion vectors respectively for 9 nutrients.
- F_j represents the j_{th} index food item in the dataset.

Objective Function (optimal food item)

$$\hat{F} = \arg\min_{F_j} \sum_{i=1}^{9} x_{ji} \log\left(\frac{x_{ji}}{r_i}\right)$$

- It helps to identify the optimal food item(from the given dataset) to consume as the fourth meal keeping the divergence from RDA values as small as possible for the day.
- $\hat{F}$ is the Optimal Food Item as the fourth meal to consume.

Objective function (Optimal Nutrient Proportions)

$$\begin{aligned} \text{minimize} \quad & \sum_{i=1}^{9} x_i \log\left(\frac{x_i}{r_i}\right) \\ \text{subject to} \quad & x - M \preceq 0, N - x \preceq 0, \\ & \sum_{i=1}^{9} x_i = 1 \end{aligned}$$

- It finds the optimal nutrient proportions not corresponding to any food item in the dataset.
- Where $x \in \mathbb{R}^9$, and let x^* is the optimal solution for the objective function.
- Then $x^* = (x_1^*, x_2^*, .., x_9^*)$ is the optimal nutrient proportions vector, where x_i^* is the ith optimal nutrient proportion.

Tools

- Exploited features in R like-
  1. "pattern matching and replacement"
  2. "split, apply and combine"
- Additional columns like category (desserts, bread, curry etc.), breakfast, lunch etc. were added using grepl() function
- Prefernce of CVXPY (Python) over CVXR (R); less dependencies e.g., rust, fortran
- Inbuilt function: kl_divergence()

```
> consumed[,1:6] ##in proportions
                             food magnesium        iron       zinc      vit_b3      vit_b6
1                Vegetable yakhni 0.8402906 0.026241043 0.01879214 0.008639459 0.002439934
2 Fruit Punch (with fresh juices) 0.2907807 0.009276336 0.00205269 0.003705404 0.001284529
3      Pea curry (Matar ki sabzi) 0.4953050 0.019538038 0.01153743 0.014612388 0.002457248
> a[1:5,1:7] ##low obj_val food
                                          food_name   magnesium         iron         zinc
119                               Chinese fried rice 0.018990032 0.0009868101 0.0005134748 0.0006
181               Pea kofta curry (Matar kofta curry) 0.012234638 0.0005597886 0.0002880429 0.0003
184 Lotus stem kofta curry (Kamal kakdi kofta curry) 0.007800397 0.0006525418 0.0001324157 0.0002
186 Cauliflower kofta curry (Phoolgobhi kofta curry) 0.007819391 0.0003999592 0.0001279840 0.0001
187     Cabbage kofta curry (Pattagobhi kofta curry) 0.007323873 0.0003388939 0.0001138135 0.0001
> colSums(b[,2:10])+ (a[1,2:10])  ##final nutrient intake in grams
     magnesium        iron        zinc      vit_b3       vit_b6       vit_b9    vit_c        vit_
119 0.06041432 0.002532594 0.001419243 0.001716608 0.0003706827 9.291347e-05 0.055734 3.305889e-0
> rda_sub[n,]  ##rda values in grams
     age_group age_yrs weight magnesium  iron   zinc vit_b3 vit_b6  vit_b9 vit_c   vit_a   vit_d
5 Adult Women     >=18     55      0.37 0.029 0.0132  0.014 0.0024 0.00022 0.065 0.00084 1.5e-05
>
```

an instance of the unconstrained problem

Limitations

- Neglects individual variabilities such as allergies, vegetarianism, etc.
- Nutrient weights (x_{ij}) don't add up linearly. Some minerals stop others from being absorbed.
- KL-Divergence returns undefined(NaN) values when a nutrient's proportion (p_{ij}) is exactly zero.
- The consumed and the suggested optimal nutrients' amount are for 100 g of every item. A square meal weighs 450 g and a dessert less than 50 g.
- The analysis ignores total calorie intake, the recommendations might lead to either calorie deficiency or excess calorie consumption.
- The analysis assumes accurate measurements from the dataset.

Conclusion & Future Scope

- Formulated daily dietary planning as a convex optimization problem & calculated the ideal nutritional profile for the fourth meal.
- Using KL Divergence as objective function, the divergence between
  1. final meal's nutrient proportions (p)
  2. the target RDA (r)

  is minimized (bounded below & above the EAR & TUL)
- Suggests an ideal food item with the appropriate nutrient proportions.
- In the future, adding calorie intake and suitably weighed food can make the model more useful in real life.

Thank You :)